Full-Stack E-Commerce Website

L'ESSENCE PARFUMERIE

Name :

.....

student ID:

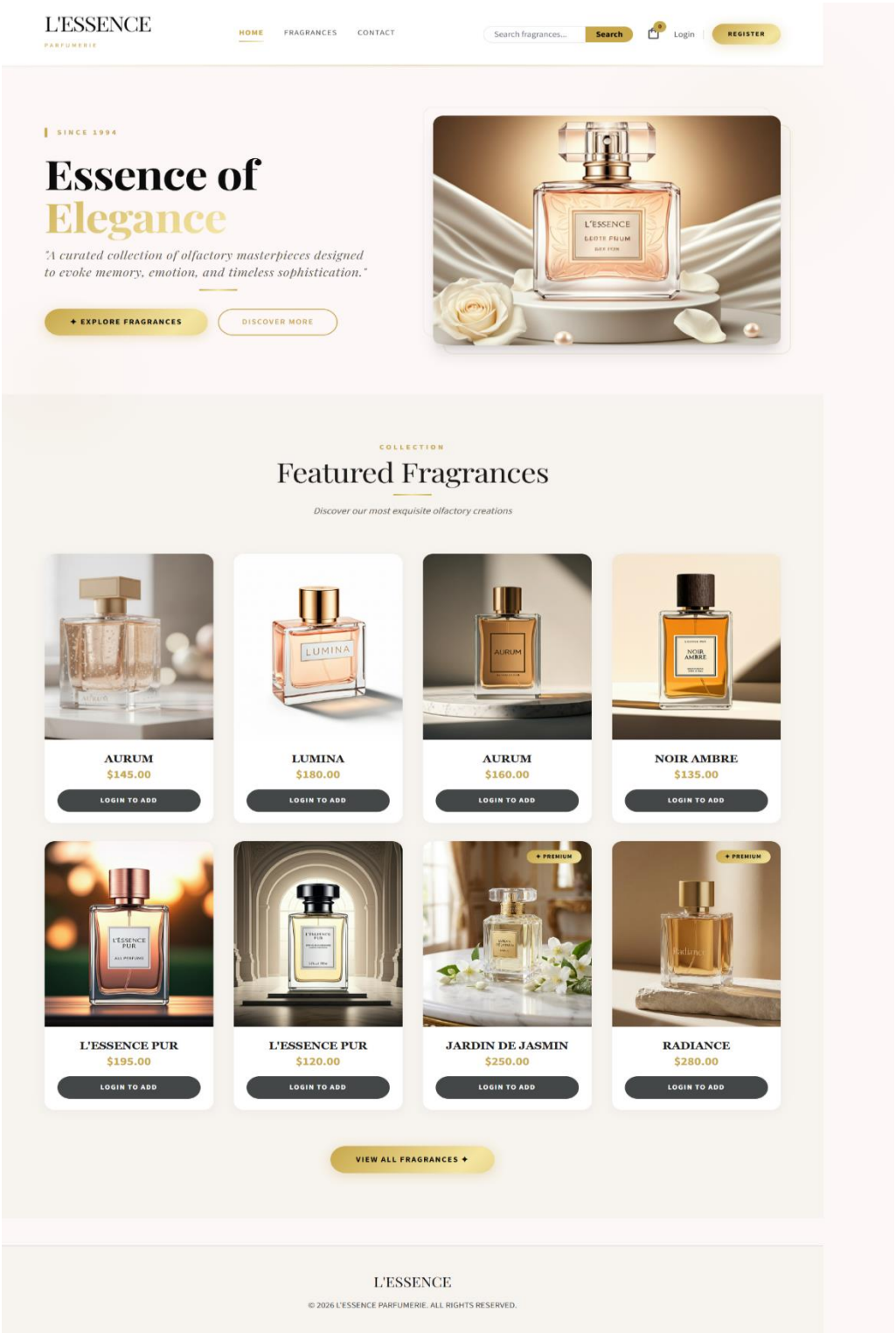
.....

Submitted to:

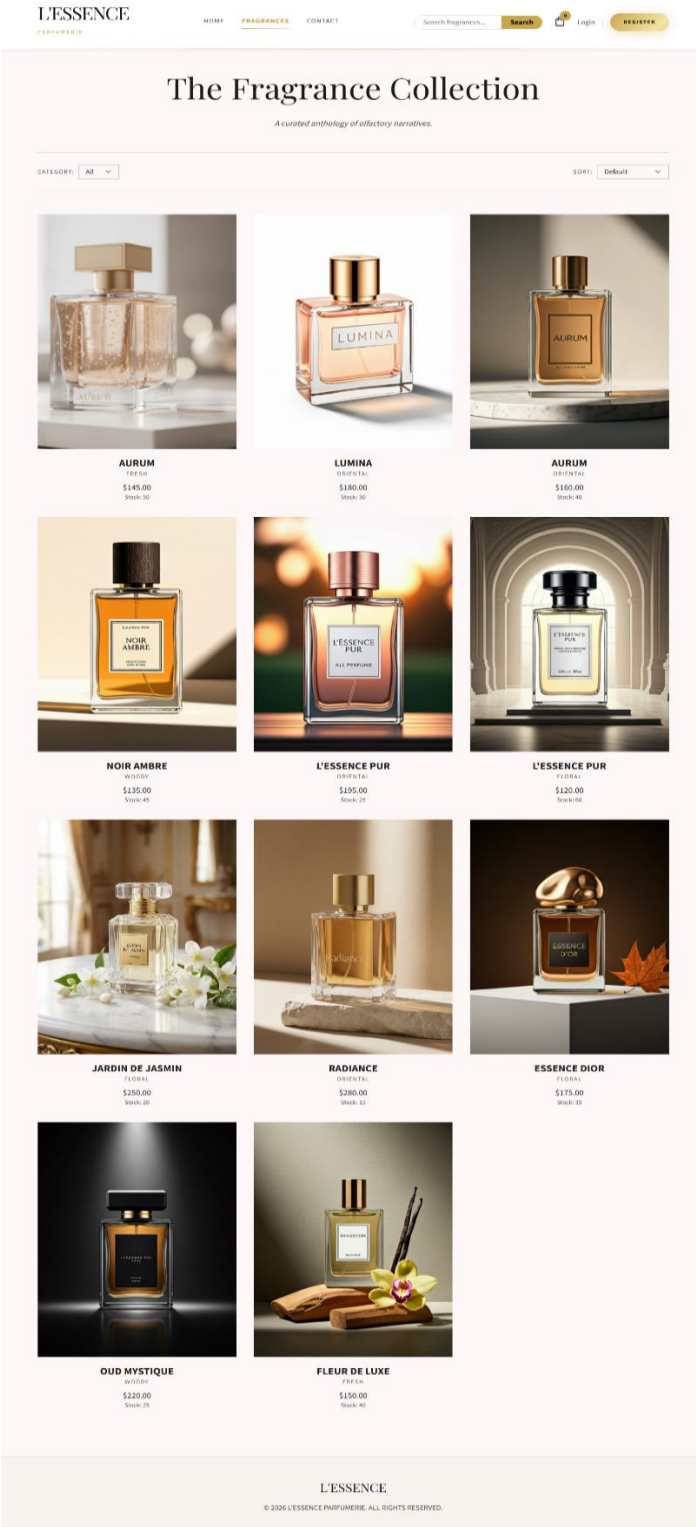
Dr.....

1. Screenshots of Each Page

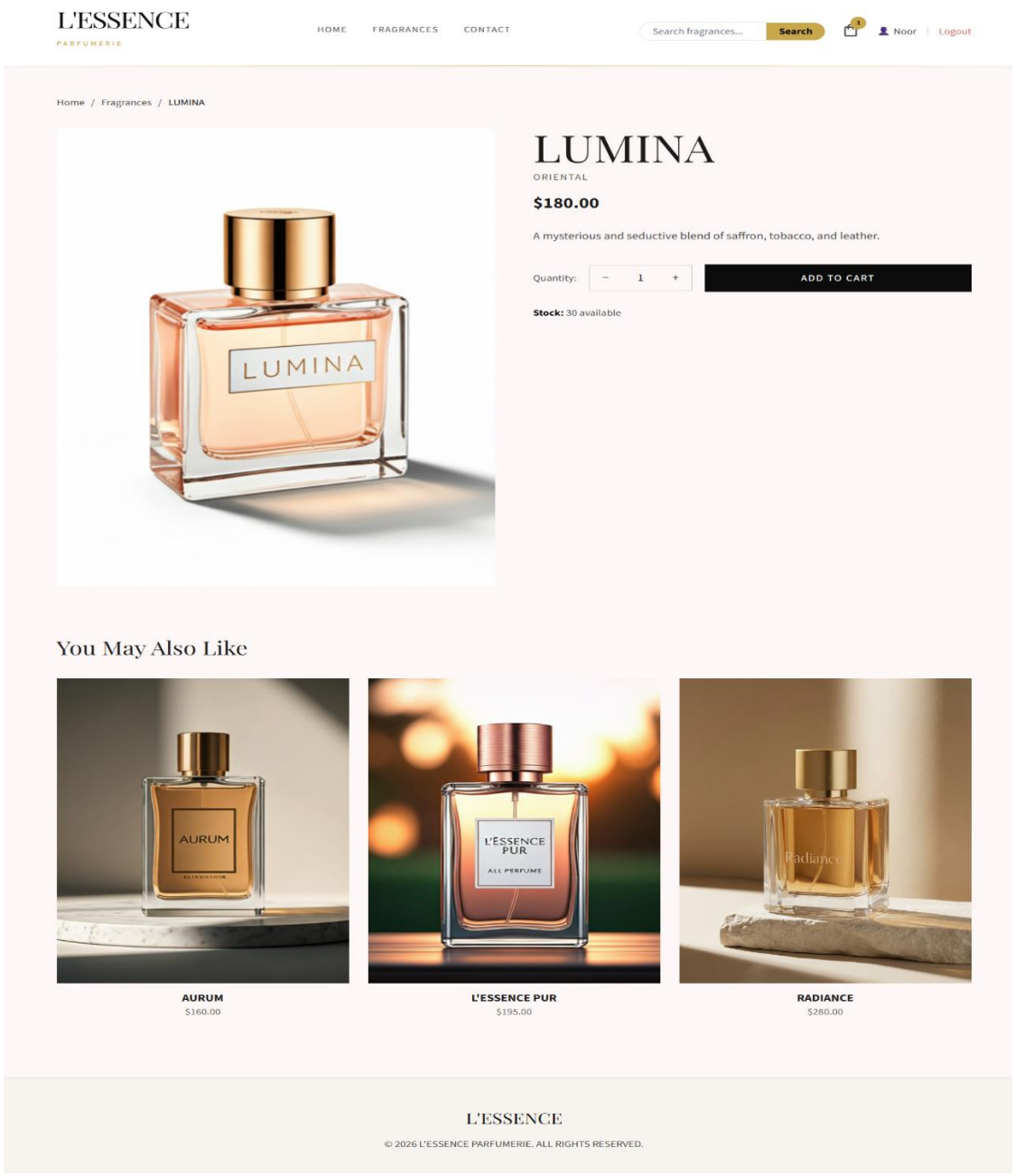
1.1 Home Page (index.php)



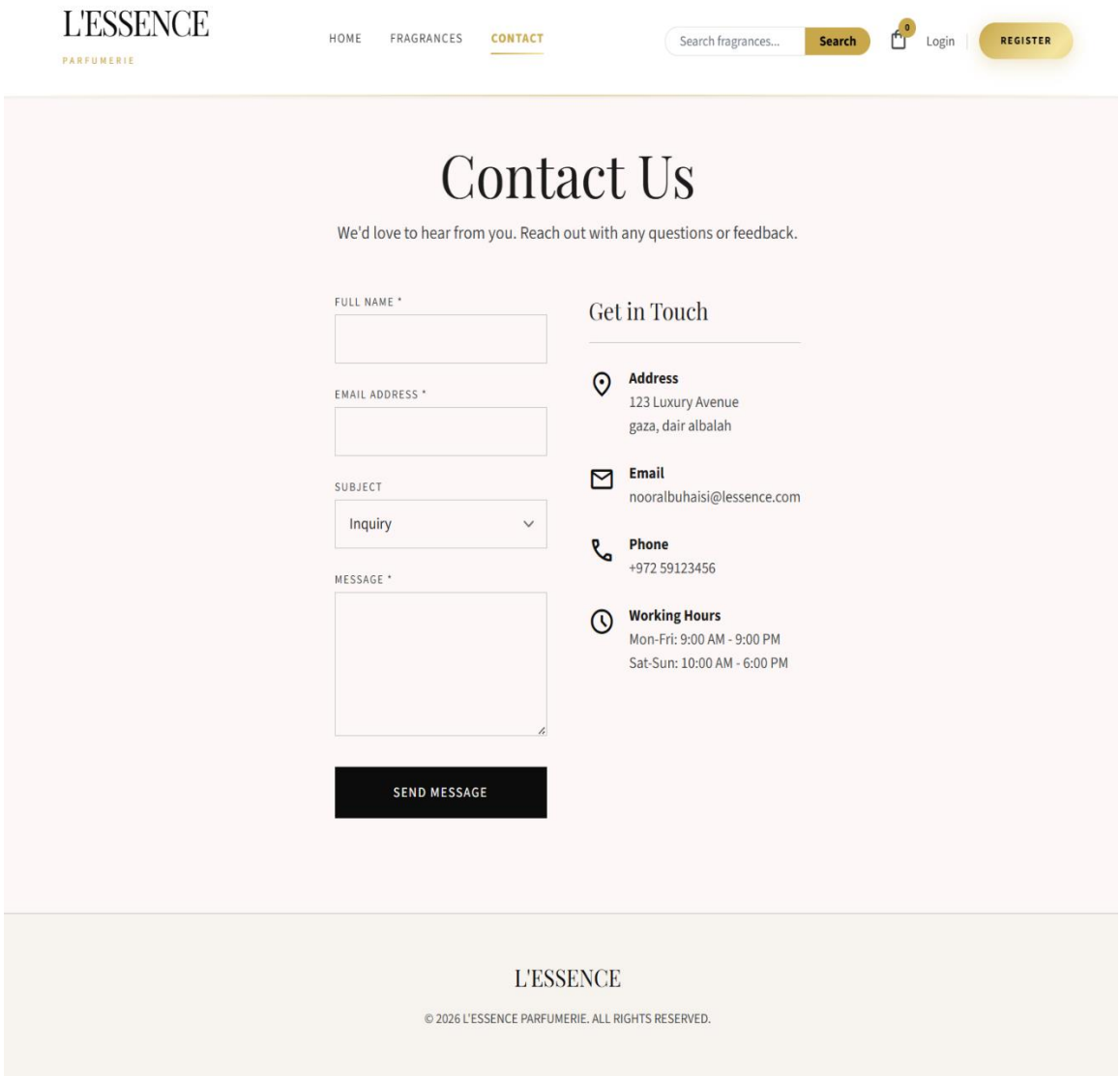
1.2 Products Page （ products.php ）



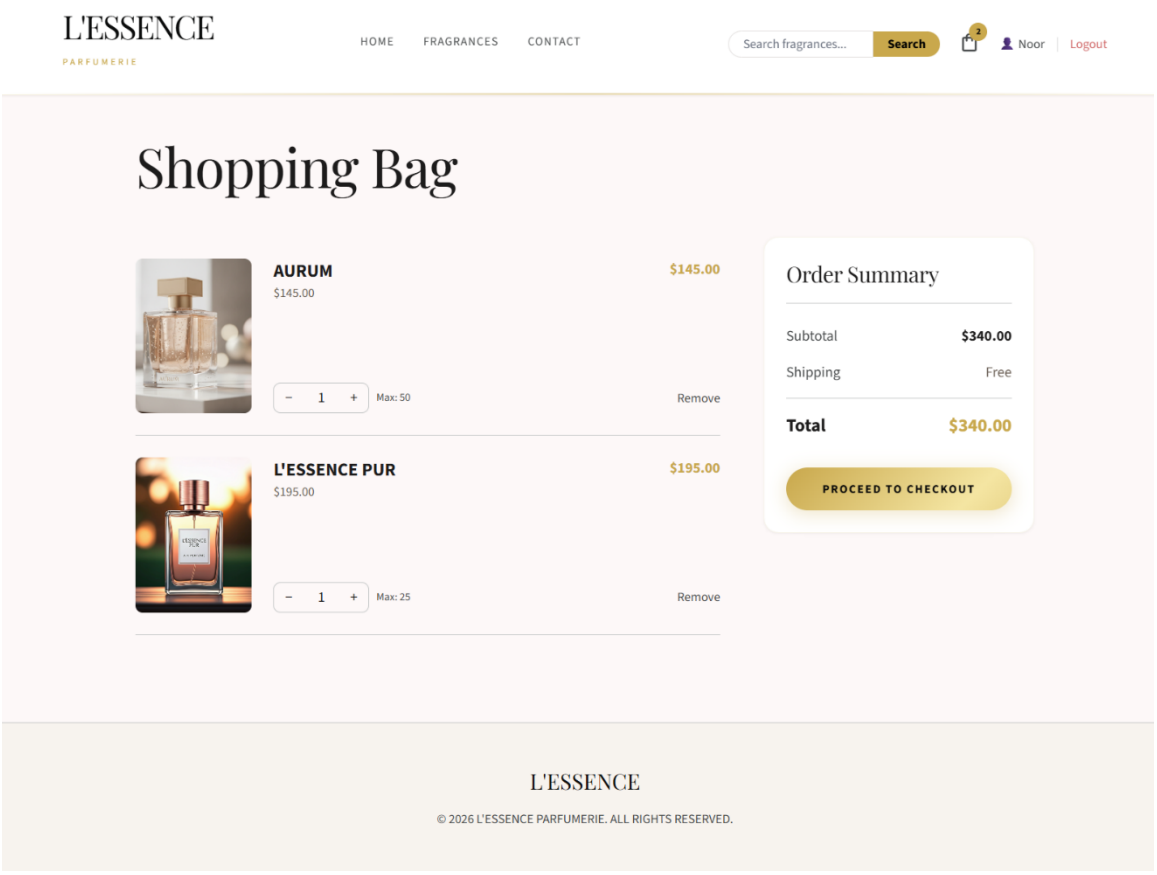
1.3 Product Detail (product_detail.php)



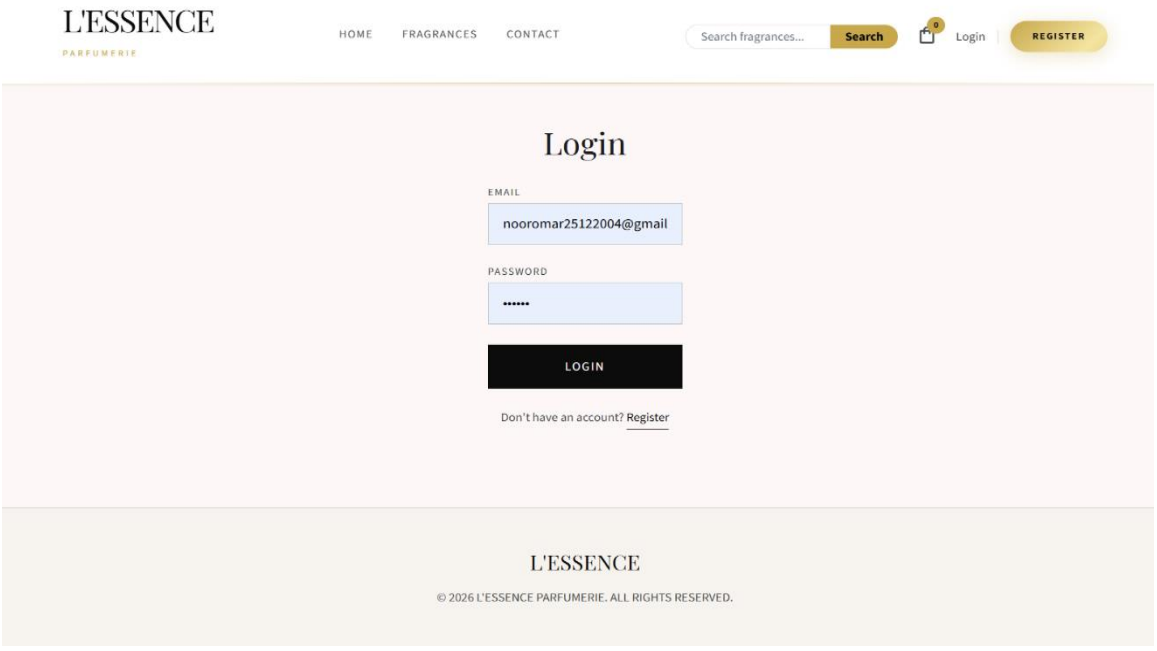
1.4 Contact Page — contact.php

- 

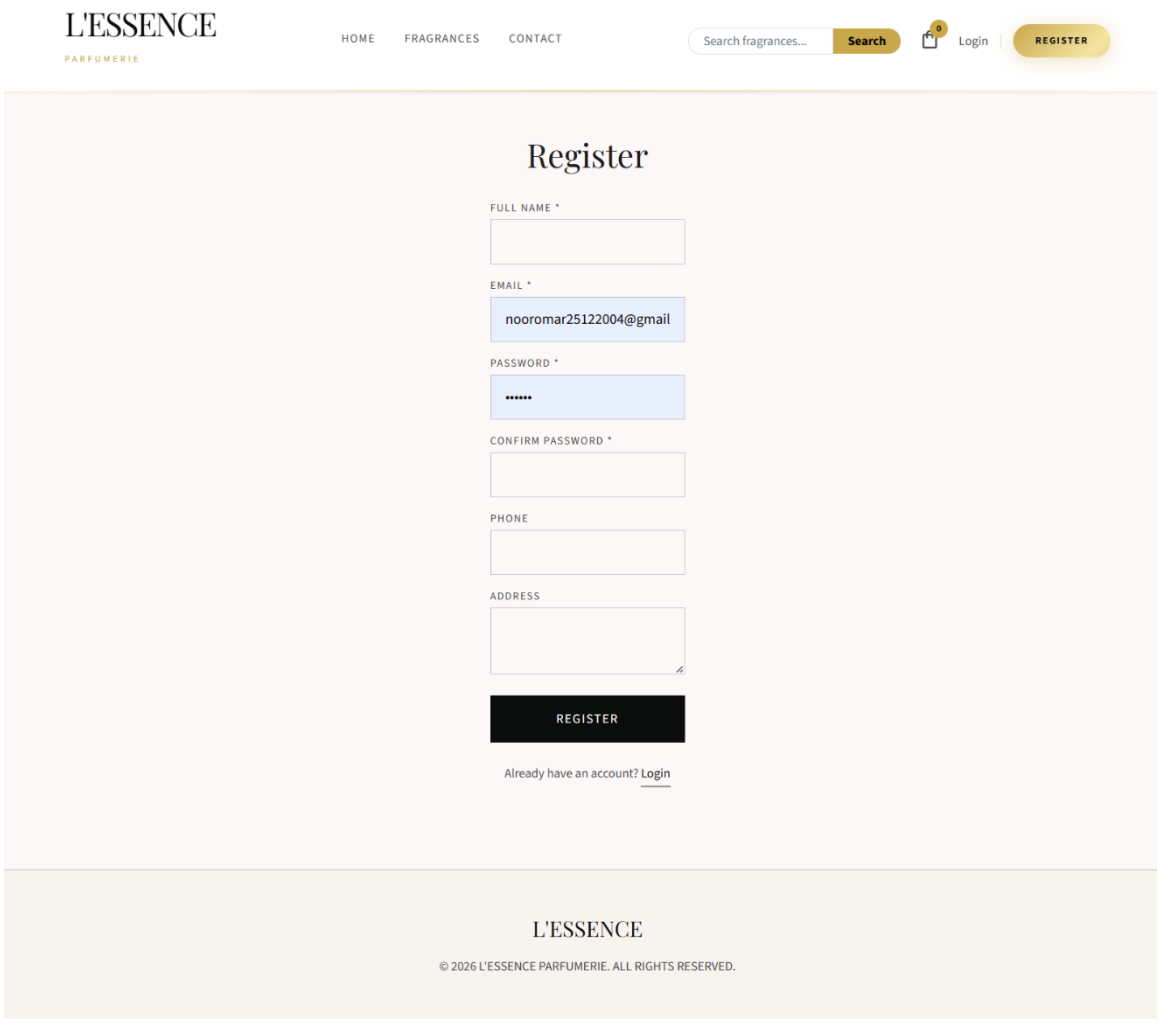
1.5 Shopping Cart (cart.php)



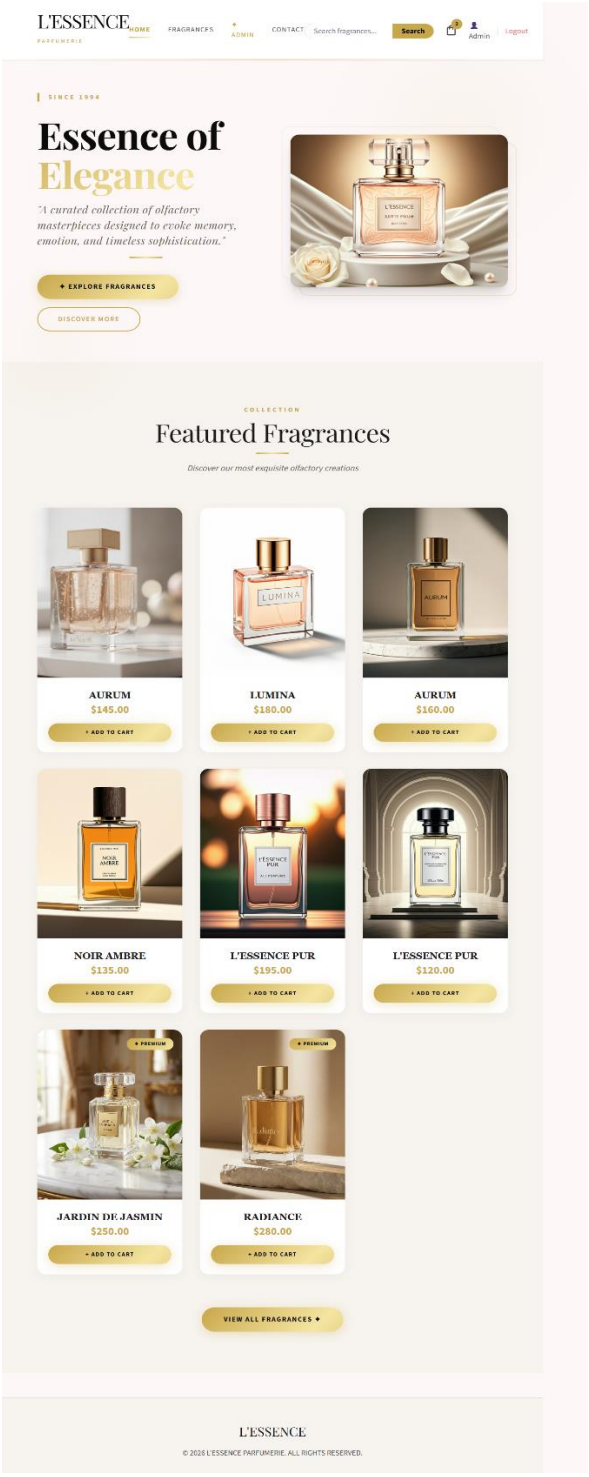
1.6 Login Page (login.php)



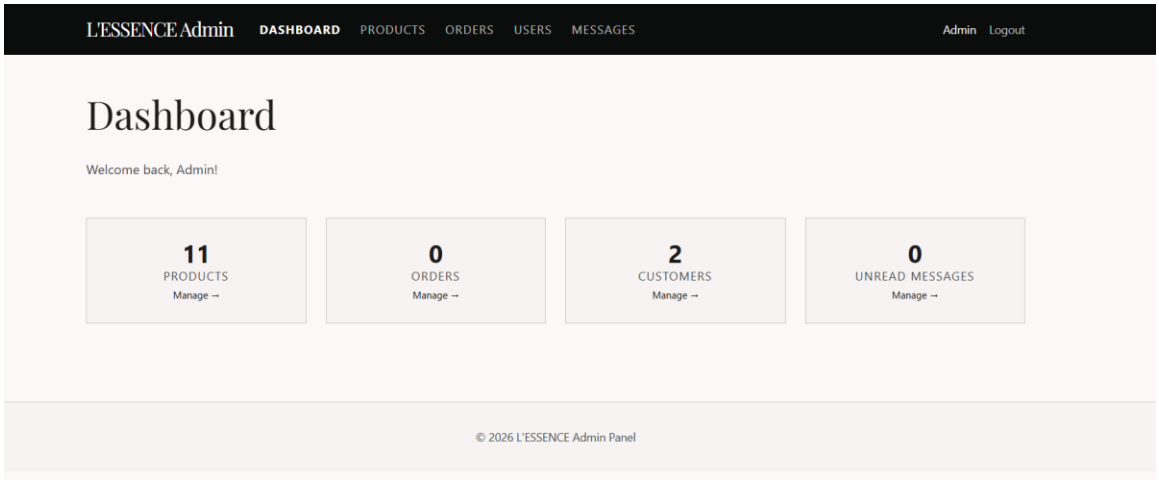
1.7 Register Page (register.php)



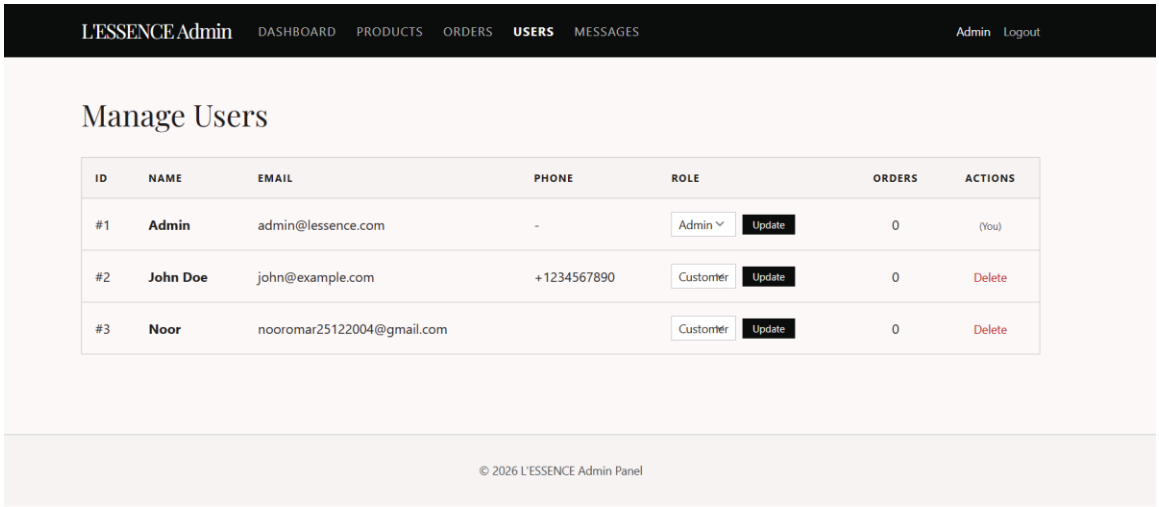
2. Admin Panel Pages



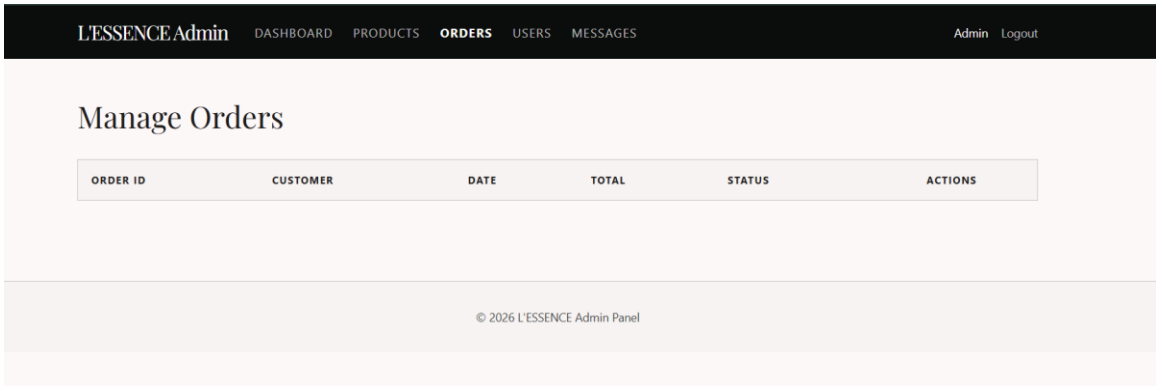
2.1 Admin Dashboard (admin/admin_dashboard.php)



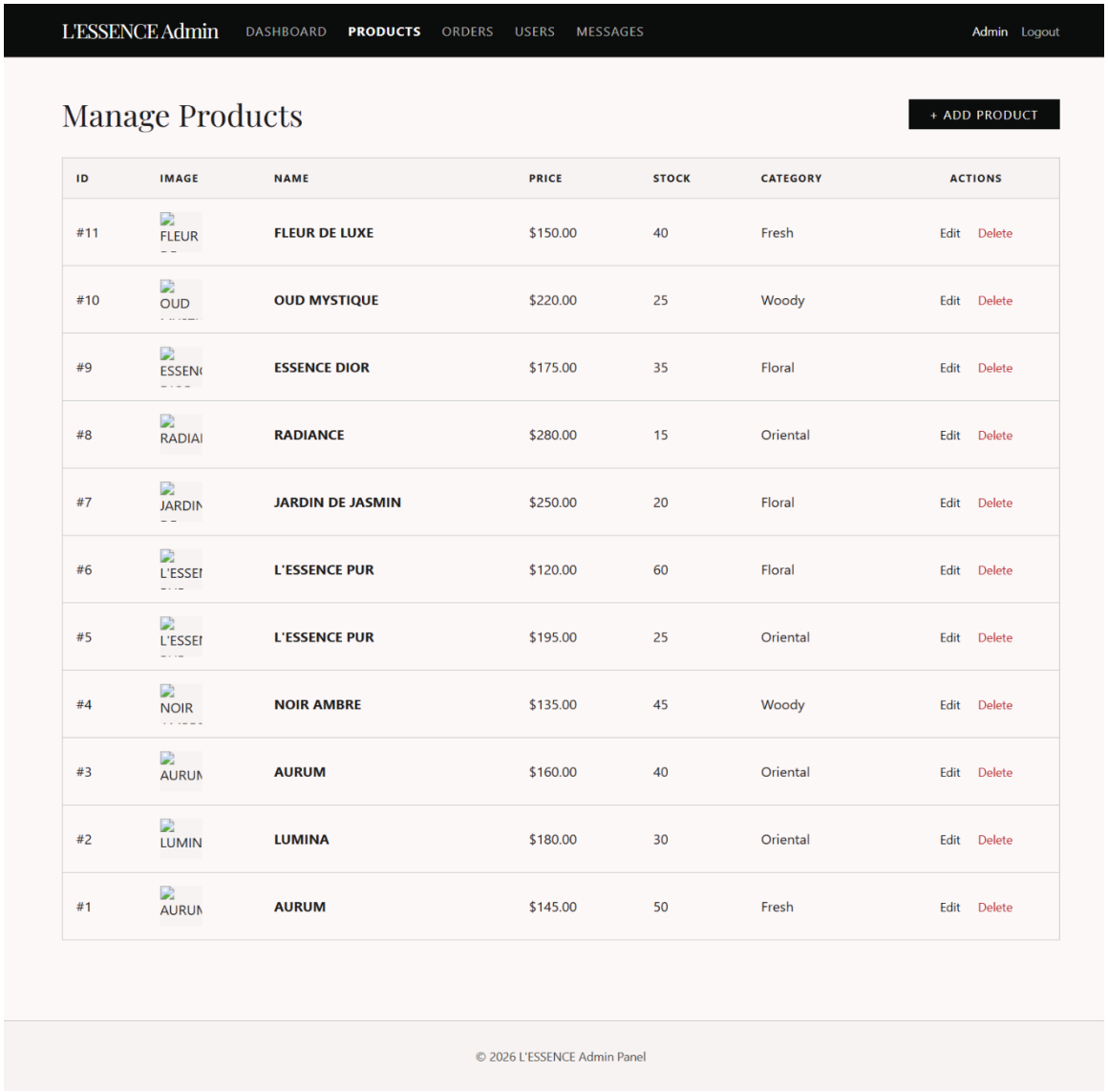
2.2 Manage users (admin/ admin_users.php)



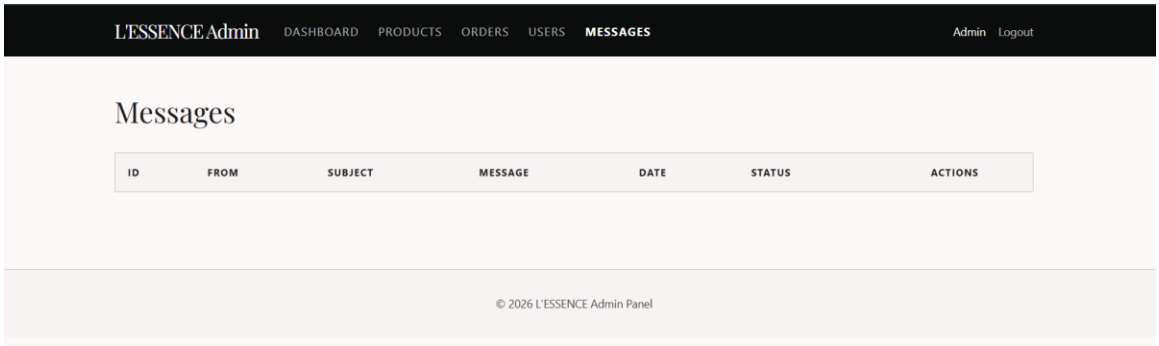
2.3 Manage Orders (admin/ admin_orders.php)



2.4 Manage Products (admin/ admin_products.php)



2.5 Manage Messages (admin/ admin_messages.php)



3. How the Front-End Communicates with the Database

The front-end pages do not access the database directly. Instead, they send requests to PHP scripts which connect to MySQL and return data.

3.1 Database Connection

config/database.php establishes a PDO connection to ecommerce_db. This file is included at the top of every PHP page.

3.2 Displaying Data (Read Operations)

Pages like index.php and products.php query the database using SELECT statements. PHP loops through the results and generates HTML (e.g., product cards). Filtering and sorting use $_GET parameters (e.g., ?category_id=2&sort=price_asc).

3.3 Sending Data (Write Operations)

Forms (login, register, contact, add to cart) use the POST method. PHP scripts read $_POST, validate inputs, and execute INSERT or UPDATE queries using prepared statements to prevent SQL injection.

3.4 AJAX Cart Interaction

Cart actions (add, update, remove) use JavaScript fetch() to send background POST requests. PHP updates the cart_items table and returns a JSON response. JavaScript updates the cart badge without reloading the page.

3.5 Sessions and Access Control

After login, login.php starts a session and stores:

- $_SESSION['user_id']
- $_SESSION['full_name']
- $_SESSION['role']

Protected pages check $_SESSION['user_id'] for authentication. Admin pages also check $_SESSION['role'] === 'admin'.

3.6 Summary of Key Data Flows

Operation	Front-End Request	PHP / Database Action	Response
Product Listing	GET request	SELECT from products + categories	HTML rendered with dynamic data
Add to Cart	POST: AJAX	INSERT/UPDATE cart_items	JSON +cart badge updates
User Login	POST	(password_verify) check	Session set; redirect based on role
Place Order	POST	INSERT into orders +order_items	Redirect to profile
Contact Form	POST	INSERT into contacts table	Success message shown to user
Admin — Status	POST	PHP UPDATE orders SET status	Table refreshed

Use Case Diagram — L'ESSENCE E-Commerce System

